

Ques. WHAT IS A MOBILE SIGNAL BOOSTER AND WHAT ARE THE DIFFERENT TYPES OF SIGNAL BOOSTERS AVAILABLE IN INDIA?

D.S.P. Rao

Ans. A signal booster amplifies a mobile phone signal sent to and from the phone to the nearby tower. A mobile phone signal booster is basically a repeater system, which takes the existing mobile phone signal, amplifies it and rebroadcasts to nearby area.

A signal booster has three main elements: outdoor or donor antenna, amplifier and indoor antenna. The donor antenna sends and receives signals from the tower, whereas an indoor antenna distributes the signal to and from a mobile phone. The amplifier, sometimes called a bi-directional amplifier (BDA) or booster, amplifies signals from the tower to the phone and the signals from phone to the tower. The amplifier is connected to the donor and indoor antennae by a coaxial cable.

A mobile phone signal booster can be of two types: home signal booster and vehicle signal booster. Home signal boosters are designed to be used in a fixed location, at home or office, while vehicle signal boosters are designed for use in a vehicle or car.

Selection of signal booster depends on frequency, type of supported network, amplifier and antenna. Mobile phone signal boosters are available for GSM frequency bands and also 2G, 3G and 4G networks. Select a booster that supports GSM, 2G, 3G and 4G networks covering all frequencies. Also, check whether the signal booster is approved by a regulatory body such as the government of India.

There are many suppliers available in India such as Mobile Boosters India, Ava Systems and Ultramind Technologies India. Depending on the brand and type, the average price of a signal booster falls between ₹ 13,000 and ₹ 22,000 for 3G support and around ₹ 30,000 for 4G support.

Q2. WHAT IS THE DIFFERENCE BETWEEN LabVIEW AND MATLAB?

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A2. LabVIEW, or Laboratory Virtual Instrument Engineering Workbench, is a system design platform and development environment for graphical language programming. It is a visual programming language from National Instruments (NI). LabVIEW programs or virtual instruments (VIs) have front panels and block diagrams. The front panel is the user interface (UI), and the block diagram contains the graphical source code, which is the programming behind the UI.

After building the front panel, the code is added to the block diagram using graphical representations of functions to control front-panel objects. The code on the block diagram is graphical code, also known as G code or block diagram code. The front panel is built using controls (inputs) and indicators (outputs). Controls supply information to the VI, and indicators indicate or display the results based on the inputs given to the VI.

LabVIEW is commonly used for data acquisition, instrument control and industrial automation on different operating systems. Graphical programming helps build programs just by dragging and dropping virtual representations of lab equipment. LabVIEW programming environment makes it easy to create small applications. For a complex code, it is important to have extensive knowledge of the special LabVIEW syntax and topology of its memory management.

MATLAB, or MATrix LABoratory, is a multi-paradigm numerical computing environment developed by Mathworks. It allows matrix manipulators, plotting of functions and data, implementations of algorithms, creation of UIs and interfacing with text-based programs written in various languages such as C, C++, Java and Python.

MATLAB supports developing applications with GUIDE, or graphical user

interface (GUI) development environment, for graphically designing GUIs. It has additional toolboxes like MuPAD symbolic engine, which allows access to symbolic computing abilities. It also has tightly integrated graph-plotting features.

Both platforms are widely used and help engineers and scientists around the world in various stages of design, simulation, modelling, prototype testing or deployment of new technologies. However, these have some differences as listed below:

1. Developers of LabVIEW are NI, whereas developers of MATLAB are Mathworks.

2. LabVIEW is mainly a graphical programming language for visualisation and parameter tuning for remote operation; MATLAB is a computing language for control algorithm development and simulation.

3. LabVIEW supports Windows, Linux, macOS and cross-platform, while MATLAB supports Windows, Linux and macOS.

4. MATLAB mainly provides mathematical/numerical computing environment, whereas LabVIEW is a system design platform that allows data acquisition, test automation, instrument control and embedded system design.

5. MATLAB is used for simulations because of additional libraries that contain higher-level functions. LabVIEW is used where a functional and intuitive GUI or interaction with hardware is required.

6. File extension for LabVIEW is .vi, and for MATLAB it is .m

7. The latest LabVIEW versions available are LabVIEW 2017 SP1 and LabVIEW NXG 2.0, released in January 2018. MATLAB's latest version available is matlabR2018a, released on March 15, 2018.

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